Local Evidence Is Not Completion: Evidence-Condition Geometry for Bounded Completion-Audit Workflows

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Abstract

Completion-audit workflows often stop before the remaining workload is known. Their stop signals–no-new findings, agent agreement, stable summaries, or self-reported completion–can be locally meaningful while still being unsafe as scope-wide completion certificates. The missing question is: under what evidence condition was the stop signal produced, and does that condition match the claimed scope? We study this certificate mismatch in bounded source-route audit workflows. A source-route stratum records both where the workflow searched and which audit route it applied. We use the resulting exposure geometry to build an evidence-condition controller with four modules: source-route exposure estimation, an eligibility gate, residual-evidence repair over weak plausible gaps, and a final SAFE/CONTINUE/ABSTAIN decision rule. Across four bounded completion-audit task families, homogeneous route reuse creates false certification risk; source-route granularity detects this risk where source-only control does not; broad eligibility alone is insufficient on the `urllib3` boundary; and the full controller avoids unsafe stops without becoming a never-stop rule. The result is a bounded diagnostic and control principle: completion certificates should be judged by the evidence condition that produced them, not by stop signals alone.

1 Introduction

This paper evaluates bounded completion-audit workflows with declared source-route scopes. These are workflows whose intended evidence region is bounded, such as a repository snapshot or document set, and whose completion claim is stated over a declared set of sources and audit routes. Examples include searching a repository snapshot for all instances of a behavior, auditing a document set for policy exceptions, checking whether an implementation covers declared failure-mode routes, or deciding whether a multi-agent audit has exhausted useful evidence within its declared scope. Such workflows must eventually stop. The difficulty is that the absence of newly found evidence is not the same object as evidence that the intended bounded workload is complete.

The failure studied in this paper is *certificate mismatch*. A workflow may produce a plausible local stop signal: repeated scans return no new findings, agents agree with each other, subtasks appear closed, or a summary becomes stable. These signals are conditioned on the parts of the task that were searched and on the routes by which they were searched. If the workflow repeatedly applies the same route to the same subset of sources, a no-new signal may certify only local exhaustion. It does not automatically support a global claim that the task is complete over its intended scope.

Consider a code-audit workflow. Auditing a file for timeout behavior does not certify that the same file has been audited for TLS behavior, exception behavior, retry behavior, or cleanup behavior. The source may be familiar, the agents may agree, and the workflow may have stopped finding timeout-related evidence; none of that establishes that other routes over the same source have been exhausted. The stop evidence is local in its evidence condition, while the completion claim is global in scope.

We formalize this observation with *evidence-condition geometry*. The basic unit is a source-route stratum, consisting of a bounded evidence region and an audit or search route. At runtime, the workflow accumulates exposure counts over these strata. The normalized exposure distribution describes the condition under which completion evidence was produced. When this distribution is localized, stop evidence is locally conditioned. It may be useful, but it is not a global completion certificate.

Based on this geometry, we define an evidence-condition controller. At a proposed stop, the controller computes the source-route exposure condition, checks whether the condition is broad enough to make the completion claim eligible for certification, and triggers repair over weak plausible strata when necessary. The controller is intentionally conservative: broad exposure gives completion eligibility, not automatic safety. A SAFE decision additionally requires that repair or audit reveal no residual evidence. If residual evidence appears, the controller outputs CONTINUE; if budget is exhausted without a defensible certificate, it outputs ABSTAIN.

Our experiments evaluate this controller on four bounded completion-audit task families: two generated scored tasks and two external real-repository audits with pattern-defined oracles. These tasks are not presented as a benchmark for the best item discovery policy. Instead, they test whether source-route exposure localization diagnoses unsafe stop claims and whether a controller can avoid accepting locally conditioned evidence as global completion proof.

This paper makes three contributions. First, it formulates false completion as a certificate mismatch between the declared scope of a stop claim and the source-route condition under which supporting evidence was produced. Second, it defines a certificate semantics for bounded audit completion:

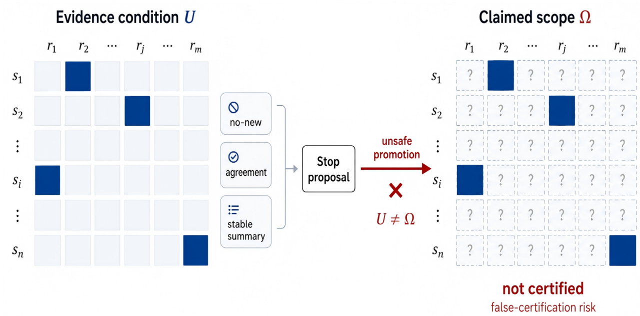

Figure 1: Certificate mismatch from local stop evidence. A workflow may observe no-new findings, agreement, or a stable summary under evidence condition U. Promoting that stop proposal to a completion claim over Ω is unsafe when $U \neq \Omega$; the unobserved source-route strata are not certified.

a SAFE decision is admissible only when the source-route evidence condition is eligible and residual-evidence repair is nonproductive under runtime-visible signals. Third, it provides bounded audit evaluation, including generated audits and external repository audits, showing that homogeneous route reuse creates false certification risk while broader source-route evidence improves eligibility without being sufficient for safety.

We deliberately do not claim that residual-potential is an optimal repair method. It is one mechanism-aligned repair instance used to test whether weak source-route regions can expose residual evidence. The main claim is about the conditions under which completion evidence can support a stop decision.

2 Related Work

High-recall review, active search, and missing mass. Technology-assisted review and total-recall systems study how to retrieve nearly all relevant items under review budgets [Grossman and Cormack, 2011, Cormack and Grossman, 2014, 2015, 2016, Grossman et al., 2016, Roegiest et al., 2015], while active search and active learning choose actions or examples to expose positives efficiently [Lewis and Gale, 1994, Settles, 2009, Garnett et al., 2012, Attenberg et al., 2015]. Missing-mass and unseen-species estimators ask how much probability mass remains unseen [Good, 1953, Chao, 1984, Chao et al., 2014]. Our scored subclasses use recall only for post-hoc false-certification measurement. The controller does not observe oracle totals, recall, or hidden missing mass; it observes the source-route condition under which a stop claim was produced and whether repair reveals residual evidence.

Agent workflows and verification. Tool-using, browser-assisted, multi-agent, and memory-based language agents allocate work across tools, agents, searches, and intermediate states [Yao et al., 2023, Schick et al., 2023, Nakano et al., 2021, Qin et al., 2024, Wu et al., 2023, Park et al., 2023, Shinn et al., 2024, Madaan et al., 2023, Wang et al., 2024]. Agent benchmarks and factual verification benchmarks evaluate task success, tool use, factuality, or evidence support [Yao et al., 2022, Deng et al., 2024, Zhou et al., 2024, Liu et al., 2024, Jimenez et al., 2024, Hendrycks et al., 2021a, Liang et al., 2022, Srivastava et al., 2023, Lin et al., 2022, Thorne et al., 2018, Gao et al., 2023, Min et al., 2023, Schuster et al., 2021]. Our question is narrower and precedes final answer scoring: whether the evidence condition under which a workflow stopped is broad enough for the scope of its completion claim.

Verify-gated completion and admission control. Verifier-gated and admission-control pipelines accept an output only after a checker, verifier, or guard passes; otherwise they reject, revise, or abstain. This fail-closed pattern is useful for completion control and is close in spirit to our SAFE/CONTINUE/ABSTAIN interface. The distinction is the certificate object. A generic verifier gate can say whether an unresolved warning remains, but it need not record *where* and *by which route* the stop evidence was produced. Our controller treats source-route evidence condition as the audited object, separates eligibility from proof, and uses residual-evidence repair to identify the next weak plausible strata rather than only rejecting a claim.

Agreement, audit agents, and oversight. Ensembles, self-consistency, verifier-guided systems, and audit agents can improve reliability [Dietterich, 2000, Cobbe et al., 2021, Wang et al., 2023]. Agreement is not useless, but it is unsafe as an unconditioned completion certificate: agents may agree because they repeatedly examined the same local region. Safety and oversight work studies broader failure modes and scalable supervision [Amodei et al., 2016, Christiano et al., 2017, Irving et al., 2018, Leike et al., 2018, Hendrycks et al., 2021b]. This paper targets a specific operational failure within that broader landscape: a workflow may present a global completion certificate whose supporting evidence was produced under a local source-route condition.

Coverage and stopping criteria. Software testing, audit coverage, and review stopping rules ask whether enough program structure, test paths, documents, or positives have been covered. Our use of source-route coverage is different: coverage is not the objective being maximized, but a diagnostic for certificate eligibility. We do not estimate how many items remain unseen, nor do we optimize active search for discovery yield. Instead, we ask whether the evidence condition that produced a stop certificate matches the scope that the certificate claims to cover.

3 Problem Setting

We study bounded completion-audit workflows with unknown residual workload and declared source-route scope. A workflow receives a task, allocates work across agents or tools, gathers evidence under a finite budget, and must decide whether the task is complete within that declared scope. Let

$$W = (T, S, R, A, B),$$

where T is the task, S is a set of sources, R is a set of routes, A is the set of agents or tools, and B is a finite budget. A source is a bounded evidence region, such as a file, module, document, source family, or repository component. A route is an audit or search lens applied to a source.

At runtime, the workflow produces evidence events

$$e_t = (a_t, x_t, r_t, u_t, o_t, c_t),$$

where a_t is the agent or tool, $x_t \in S$ is the source, $r_t \in R$ is the route, u_t is the action, o_t is the observation, and c_t is the cost. Every evidence event is produced under a source-route condition. A source-route stratum is $s = (x, r)$, and the intended evidence-condition space is

$$\Omega = S \times R.$$

The source-route space does not require the workflow to know all true missing items in advance. It requires the intended audit scope to be defined: which sources and routes are relevant to the completion claim. This assumption is weaker than knowing the answer set and stronger than an unbounded open-world search. It matches many practical audits: a system may know which repositories, documents, modules, or evidence sources are in scope, and it may know which audit routes are relevant, while still not knowing how many true findings remain. The controller therefore evaluates the condition under which evidence was gathered, not the unknown total workload itself.

A stop claim is a workflow assertion

$$C_t : \quad T \text{ is complete over the intended workload scope.}$$

A false certification occurs when the workflow accepts SAFE for a stop claim that is not supported by the evaluation oracle:

$$\text{SAFE}(C_t) = \text{true}, \qquad \text{complete}(T) = \text{false}.$$

In item-discovery subclasses, this is operationalized as accepting SAFE while bounded-oracle recall is below the completion threshold. In broader completion audits, it means accepting a global completion claim while relevant support, contradiction, or unresolved evidence remains outside the evidence condition. The oracle is used only for evaluation. It is not available to route assignment, repair scoring, eligibility checks, or stop decisions.

4 Evidence-Condition Geometry

Let $v_t(s)$ be the runtime-visible exposure count for source-route stratum s up to time t. Exposure may count visits, scans, route-specific audits, tool calls, or other logged search actions. Define

$$p_{\text{exp}}(t, s) = \frac{v_t(s)}{\sum_{s' \in \Omega} v_t(s')}.$$

This distribution is a point on the source-route coverage simplex. It describes where and how completion evidence was produced.

We use two runtime summaries:

$$\text{support}(t) = \frac{|\{s : v_t(s) > 0\}|}{|\Omega|},$$

$$G_{\text{exp}}(t) = \text{Gini}(\{v_t(s) : s \in \Omega\}).$$

These summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

Proposition 1 (Local no-new evidence is not a global certificate)**.** *Let $U \subset \Omega$ be a strict subset of the intended source-route space. Suppose all evidence used to support a stop claim C_t is produced inside U, and suppose the workflow observes no-new evidence only for actions inside U. Then the no-new evidence can support local exhaustion over U, but it cannot by itself support global completion over Ω.*

Proof sketch. No-new evidence is conditioned on the source-route actions that were executed. If no events are produced in $\Omega \setminus U$, the workflow has not observed the absence of residual evidence in those unexposed strata. A world in which U is exhausted and $\Omega \setminus U$ contains residual evidence is observationally indistinguishable from a globally complete world under the workflow's local evidence log. Therefore local no-new evidence rules out additional findings only under the searched condition U; without additional assumptions about unsearched strata, it cannot rule out residual evidence in $\Omega \setminus U$.

Source-only geometry is too coarse: auditing a file for timeout behavior does not certify that TLS, retry, exception, or cleanup behavior has been audited in that file. Source-route-action geometry is more detailed, but it is sensitive to logging conventions and current experiments do not show a stable advantage over source-route. Source-route is the default because it captures both where and how evidence was produced while remaining interpretable and runtime-computable. The geometry is thus a certificate diagnostic rather than a semantic theory of all possible evidence. It asks whether the current log covers the intended source-route space broadly enough for the proposed stop claim; it does not assert that all routes are equally important or that support/Gini thresholds transfer unchanged across domains.

5 Method

The method has four modules: source-route exposure estimation, eligibility gating, residual-evidence repair, and the final SAFE/CONTINUE/ABSTAIN decision rule. The point is not to prove universal completion; it is to make the runtime evidence condition explicit before a stop claim is accepted.

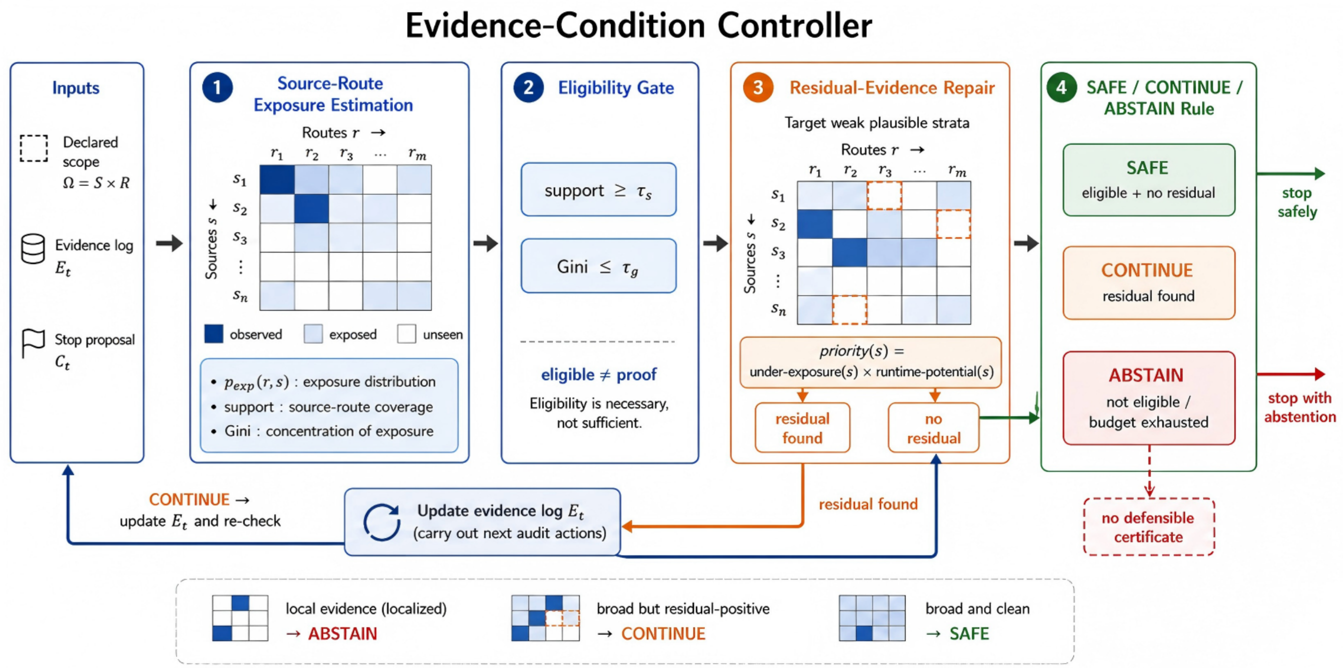

Figure 2: Evidence-condition controller. The controller maps a runtime evidence log, declared source-route scope, and stop proposal to an exposure distribution over the intended source-route space. The support/Gini check is an eligibility test, not a completion proof. If the condition is localized or weak plausible gaps remain, the controller repairs or continues. It returns SAFE only when the evidence condition is broad and repair or audit reveals no residual evidence; otherwise it returns CONTINUE or ABSTAIN.

5.1 Source-Route Exposure Estimation

Input: a runtime evidence log E_t, intended source set S, intended route set R, and their product space $\Omega = S \times R$. Output: exposure counts $v_t(s)$, normalized exposure $p_{\text{exp}}(t, s)$, source-route support $\text{support}(t)$, and exposure concentration $G_{\text{exp}}(t)$. Role: summarize where and how evidence was gathered, so localized stop evidence can be distinguished from broad evidence. Algorithm 1 shows how the four modules are composed at a proposed stop.

The source-route exposure summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

5.2 Eligibility Gate

Input: $v_t(s)$, $p_{\text{exp}}(t, s)$, $\text{support}(t)$, $G_{\text{exp}}(t)$, thresholds τ_s and τ_g. Output: eligible / ineligible. Role: decide whether the current evidence condition is broad enough for a stop claim to be considered.

The eligibility gate is not a completion proof. It is only a test of whether the evidence condition is broad enough for a global stop claim to be considered. In the current implementation, support and Gini thresholds instantiate this operational gate; they are not claimed as universal laws.

5.3 Residual-Evidence Repair

Input: current evidence log E_t, evidence condition Ω, claim C_t, budget B, and repair rule $\mathcal{R}$. Output: repaired log or continued audit targets. Role: probe weak plausible strata for residual evidence when eligibility alone is insufficient. Weak plausible gaps are strata that are both under-exposed in the current source-route condition and still plausible under non-oracle runtime signals such as source text, route names, route match counts, source size, or ledger-visible agent notes.

The repair instance used in the experiments is

$$\text{priority}(s) = \text{under\_exposure}(s) \cdot \text{runtime\_potential}(s).$$

The under-exposure term gives priority to weak parts of the current completion certificate. The runtime-potential term uses visible information such as source text, route names, source size, route match counts, or non-oracle ledger signals. It is not allowed to use oracle labels, oracle totals, oracle missing mass, undiscovered true item counts, post-hoc recall, or scorer-visible target distributions. Residual-potential is a mechanism-aligned repair instance, not an optimal active-search algorithm.

5.4 SAFE/CONTINUE/ABSTAIN Decision Rule

Input: eligibility result and residual-evidence repair outcome. Output: SAFE, CONTINUE, or ABSTAIN. Role: accept a stop only when the evidence condition is broad and repair or audit reveals no residual evidence.

Algorithm 1 Evidence-Condition Controller

Require: log E_t, space Ω, claim C_t, budget B, thresholds τ_s, τ_g, repair rule $\mathcal{R}$
Ensure: decision in $\{$SAFE, CONTINUE, ABSTAIN$\}$
Compute $v_t(s)$, $p_{\exp}(t,s)$, $\mathrm{support}(t)$, and $G_{\exp}(t)$
$eligible \leftarrow [\mathrm{support}(t) \geq \tau_s \wedge G_{\exp}(t) \leq \tau_g]$
if not $eligible$ and $B = 0$ **then**
  return ABSTAIN
end if
if not $eligible$ or weak plausible gaps remain **then**
  Audit $Q = \mathcal{R}(E_t, \Omega, C_t)$ within B
  if residual evidence appears **then**
    return CONTINUE
  end if
  Recompute $eligible$
end if
if $eligible$ and no residual evidence appeared **then**
  return SAFE
end if
return ABSTAIN

The final rule is intentionally conservative. Broad exposure gives completion eligibility, not automatic safety. A SAFE decision additionally requires that repair or audit reveal no residual evidence. If residual evidence appears, the controller outputs CONTINUE; if budget is exhausted without a defensible certificate, it outputs ABSTAIN.

Proposition 2 (Controller certificate semantics)**.** *For a declared source-route scope Ω, thresholds τ_s, τ_g, and a fixed repair rule $\mathcal{R}$ that uses only runtime-visible signals, Algorithm 1 returns* SAFE *only if (i) the exposure condition satisfies the eligibility gate and (ii) repair or audit reveals no residual evidence within the allocated repair budget. If residual evidence is found, the returned decision is* CONTINUE*; if eligibility cannot be established under budget, the returned decision is* ABSTAIN.

Proof sketch. The algorithm computes eligibility before any SAFE branch is reachable. Ineligible states either enter repair or return ABSTAIN when no budget remains. During repair, any residual evidence triggers the CONTINUE branch. The only SAFE branch is guarded by both eligibility and the absence of residual evidence. The statement is therefore a soundness property of the controller's declared certificate semantics, not a guarantee that the world contains no unobserved evidence.

6 Experiments

The experiments test whether a proposed stop can be accepted as a bounded completion certificate under a declared source-route scope. They are not a benchmark for the best item-discovery policy. Each run follows the same closed protocol. First, a workflow trajectory produces a proposed stop state using only runtime-visible evidence. Second, the compared decision rule observes the exposure log, applies any allowed repair under the fixed budget, and returns SAFE, CONTINUE, or ABSTAIN. Third, the oracle is revealed only for post-hoc scoring. The controller is therefore evaluated on whether it accepts unsafe completion claims, not on whether it has oracle access to the remaining workload.

Generalization axes. The validation varies five axes rather than a single task instance: task family (policy documents, generated code, and two external repository audits), data source (generated hidden-oracle tasks and frozen real-package snapshots), route assignment (homogeneous reuse, route-partitioned audit, and the `urllib3` extended audit), threshold/budget sweep (τ_s, τ_g and repair budgets in the supplement), and seed/config reproduction (three-seed smoke reproduction plus the full 200-seed validation). These axes test whether local-evidence false certification recurs across bounded source-route completion-audit workflows, without claiming universal open-world completion.

We organize the evaluation around five method-aligned questions. Q1 asks whether naive stop signals create false certification. Q2 asks whether source-route geometry detects risks that source-only exposure misses. Q3 asks whether broad eligibility is sufficient for safety. Q4 asks whether the full controller rejects unsafe stops without degenerating into a never-stop rule. Q5 compares residual-potential repair with random and high-potential repair under the same decision rule and checks whether threshold/budget choices change the safety-cost tradeoff.

Tasks. `policy-docset` is a generated policy document set with four sources covering access control, data handling, release process, and audit exceptions. Its routes are obligation, exception, deadline, and prohibition; the hidden oracle contains 24 policy clauses. `code-repo` is a generated bounded code repository with authentication, payment, storage, and API-client sources. Its routes are compatibility, security, and resilience; the hidden oracle contains 20 code-risk items.

`requests` is an external real-repository audit over six files from a local installed snapshot of the Python package: adapters, API, authentication, models, sessions, and utilities. Its routes are TLS, timeout, exception, and compatibility. `urllib3` is an external real-repository audit over connection, connection-pool, pool-manager, response, retry, and timeout sources. Its routes are timeout, retry, TLS, exception, and cleanup, giving 30 intended source-route strata. External oracles are pattern-defined from frozen snapshots: stronger than generated toys, weaker than human-annotated completion-audit benchmarks.

Workflow conditions. Homogeneous route reuse assigns multiple agents to the same route across sources. The reused routes are obligation for `policy-docset`, compatibility for `code-repo`, and timeout for the two external repository audits. Route-partitioned audit assigns agents to different routes over the same source set. `urllib3` additionally includes an `extended_audit` condition covering all five routes.

Compared policies and ablations. We compare policies on fixed bounded-audit states. Decision variants are Naive stop, Source-only, a generic verifier-gate baseline, Source-route eligibility-only, and the full controller. The verifier-gate baseline does not use source-route geometry; it returns SAFE only when a runtime checker reports no unresolved warning, and otherwise returns ABSTAIN. It is a conservative admission-control baseline, not a repair controller. Repair-target variants keep the full decision rule fixed and choose strata by Random, High-potential, or Residual-potential. All policies use the same fixed trajectories and proposed stop states; differences come from the decision rule or repair target selection, not from different discovery trajectories. Oracle labels are used only for post-hoc scoring.

We name five ablations explicitly. Source-only vs. source-route is the *granularity ablation*; eligibility-only vs. full is the *proof-separation ablation*; Naive, Source-only, Verifier-gate, Eligibility-only, and Full form the *decision-rule ablation*; and Random, High-potential, and Residual-potential form the *repair-target ablation*. Varying τ_s, τ_g and repair budget gives the *threshold/budget sensitivity* ablation.

Seeds, budgets, thresholds, and cost. Generated tasks use repair budget 4; `requests` and `urllib3` use 200 seeded validations with budgets 4 and 5, respectively. The controller uses $\tau_s = 0.75$, $\tau_g = 0.70$ as pre-specified operational test points, not oracle-tuned hyperparameters. A recall threshold of 0.90 is used only for bounded-oracle false-certification evaluation and is invisible to the controller. Repair cost counts only additional scanned source lines plus extraction events after a stop state is fixed. Seeds, route assignments, repair orders, and challengers are fixed before oracle scoring.

Reproducibility and leakage control. We provide both a lightweight three-seed reproduction configuration and the full seeded validation configuration. The configs specify task snapshots, route sets, thresholds, budgets, seeds, repair rules, oracle-scoring paths, output CSVs, and figure paths. Runtime decisions may use only the evidence log, source text, lexical route matches, route identifiers, source size, and non-oracle ledger notes. They may not use oracle totals, undiscovered item counts, post-hoc recall, or scorer-visible target distributions. The released scripts regenerate Table 1, Figure 3, Figure 4, and the controller-count tables from the same fixed configs.

Credibility checks. To separate diagnostic behavior from artifacts of one oracle or ordering, we add three audit checks. First, seeded safe-state perturbations vary repair order and budget on broad complete states; the desired behavior is to remain SAFE rather than over-react to extra audit opportunity. Second, route-match sanity checks sample external oracle positives by route and retain ambiguous matches as ambiguous rather than treating them as full validation. Third, scalar and source-only negative controls test whether simpler completion proxies can explain the observed decisions. These checks are reported in the supplement and are used only as credibility evidence, not as new open-world task claims.

The four task families supply bounded completion-audit cases with withheld or post-hoc oracles. Simulation sweeps are screening checks, not new task claims. Leakage controls are in the supplement; runtime decisions use visible logs and source text, not oracle labels or undiscovered true-item counts.

7 Results

The results follow a single diagnostic chain: local stop signals can be localized rather than certifying; source-only coverage can create a false sense of global evidence; source-route geometry exposes the mismatch; eligibility is only a precondition, not proof; and residual repair closes the loop by turning a candidate stop into SAFE, CONTINUE, or ABSTAIN.

Local stop signals are not certificates. Figure 3 begins with the failure mode. Across all four tasks, homogeneous route reuse produces localized evidence and would falsely certify completion if accepted as SAFE. Controlled sweeps in the supplement show the same qualitative pattern under localized exposure. This motivates source-route granularity rather than accepting unconditioned stop signals. This validates the first link in the diagnostic chain: local stop evidence can be real while still being the wrong certificate for a scope-wide claim.

Source-only hides route mismatch. The middle panel of Figure 3 is the source-only vs. source-route ablation. Homogeneous runs touch all files, so source-only support looks complete; the same states have localized source-route support and sub-threshold recall. Source-only exposure cannot detect that timeout evidence does not certify TLS, retry, exception, or cleanup routes in the same file. In the four task families, source-only support is 1.0 at the homogeneous stop, while source-route support ranges only 0.20–0.33 and bounded-oracle recall ranges 0.104–0.708. This validates the second link: source coverage alone can hide a route-conditioned mismatch.

Source-route geometry exposes mismatch. The source-route view records both where evidence was gathered and which route produced it. In homogeneous conditions, this geometry stays concentrated even when source-only support is full; in route-partitioned and extended audits, support broadens and concentration falls. The diagnostic value is therefore not that source-route metrics prove completion, but that they expose when the stop claim is conditioned on too narrow an evidence route. This validates the third link: source-route geometry turns hidden route mismatch into an observable stop-time warning signal.

Eligibility is only a precondition. Route-partitioned evidence improves eligibility, but it is not proof. In `urllib3`, route-partitioned exposure covers 24/30 strata and passes the gate, yet recall remains below threshold and repair finds residual evidence; the controller returns CONTINUE, not SAFE.

Table 1: Experimental coverage. SR strata denote source-route strata; oracles are used only for post-hoc scoring.

Study	Sources	Routes	SR strata	Oracle items	Runs/seeds	Main measurement
`policy-docset`	4	4	16	24	200/challenger	FCR, repair gain
`code-repo`	4	3	12	20	200/challenger	FCR, repair gain
`requests`	6	4	24	298	200/challenger	controller validation
`urllib3`	6	5	30	699	200/challenger	boundary validation
Controlled geometry	–	variable	variable	synthetic	2160	metric screening
Exposure localization	–	variable	18/30/42	synthetic	2700	localization check

Table 2: Boundary case showing why eligibility is not proof. The state passes the source-route eligibility gate, but post-hoc recall remains below the completion threshold; repair-aware control therefore returns CONTINUE rather than SAFE.

Task	State	Support	Gini	Recall	Residual+	Elig.-only	Full
urllib3	route_partitioned	0.800	0.647	0.835	yes	SAFE	CONTINUE

Table 3: Seeded unsafe/complete controller decision counts for Figure 4. Unsafe and Complete are the seeded denominators; FCR is SAFE-on-unsafe divided by Unsafe, and safe coverage is SAFE-on-complete divided by Complete. These denominators differ from the one-state eligibility boundary in Table 2. Oracle labels are used only after decisions are fixed. S/C/A reports SAFE/CONTINUE/ABSTAIN counts.

Decision rule	Unsafe	Complete	Unsafe S/C/A	Complete S/C/A	FCR	Safe cov.
Naive stop	400	1200	400/0/0	1200/0/0	1.000	1.000
Source-only	400	1200	400/0/0	1200/0/0	1.000	1.000
Verifier-gate	400	1200	0/0/400	1200/0/0	0.000	1.000
Eligibility-only	400	1200	0/0/400	1200/0/0	0.000	1.000
Full	400	1200	0/400/0	1200/0/0	0.000	1.000

The extended audit reaches support and recall 1.000. Table 2 is a fixed boundary-case table: its denominator is the single eligible but residual-positive `urllib3` state, not the seeded unsafe-state denominator used for Table 3. This validates the fourth link: eligibility permits a completion claim to be considered, but does not certify it.

Residual repair closes the stop-decision loop. A naive controller falsely certifies homogeneous stops; the full controller rejects them. It is not a never-stop rule: it returns SAFE for broad complete states in the generated tasks, `requests`, and the extended `urllib3` audit. In seeded external validations, repair states are oracle-unsafe and return CONTINUE or ABSTAIN, while oracle-safe complete-state perturbations keep safe coverage 1. The generic Verifier-gate baseline also avoids false SAFE decisions when residual warnings are present, but it does so by abstaining and gives no source-route diagnosis or next audit target; the full controller instead returns CONTINUE on residual-positive unsafe states. Figure 4 therefore reports seeded unsafe-state FCR rather than relying only on the five fixed observed stop states; those fixed states define the boundary cases and are counted in the supplement. Table 3 uses seeded unsafe and complete decision counts: FCR is SAFE-on-unsafe divided by the seeded unsafe denominator, while safe coverage is SAFE-on-complete divided by the seeded complete denominator. These denominators intentionally differ from Table 2, which isolates one eligibility-only boundary state.

Repair policy comparison. Figure 4 separates repair target variants from decision variants. Residual-potential exposes residual evidence and has higher mean gain than random repair, but it is not proven optimal; high-potential is competitive in cost-normalized terms. Repair is therefore a bounded probe for remaining evidence, not a claim of optimal active search. The supplement includes per-task decision breakdowns and lightweight external baselines, including source-only stopping, random route expansion, high-potential expansion, and a scalar singleton/Chao negative control. This validates the final link: residual repair converts an eligible stop into a closed SAFE/CONTINUE/ABSTAIN decision rather than a bare metric threshold.

Small claim-verification pilot. To test whether the controller is merely a fail-closed verifier, we add a small deterministic claim-audit pilot with human-style routes: support, contradiction, exception-path, config-default, and scope-boundary. A generic verifier gate avoids unsafe SAFE decisions by abstaining whenever an unresolved warning remains, but it also abstains on one safe state and gives no repair target. The full controller uses source-route eligibility plus residual repair, avoiding false certification while preserving safe coverage in this pilot. The pilot is intentionally small and not a human-annotated benchmark; it checks interface behavior for completion control.

Sensitivity and audit checks. Sensitivity sweeps are in the supplementary material. Varying τ_s, τ_g and repair budget changes the SAFE/CONTINUE/ABSTAIN mixture and repair cost, but does not create false certifications in the external controller validations. Oracle construction, negative controls, leakage checks, seeded details, and regex examples are also supplementary.

Together, the results close the intended bounded-audit loop: naive and source-only stopping certify localized evidence, source-route geometry exposes the mismatch, eligibility is

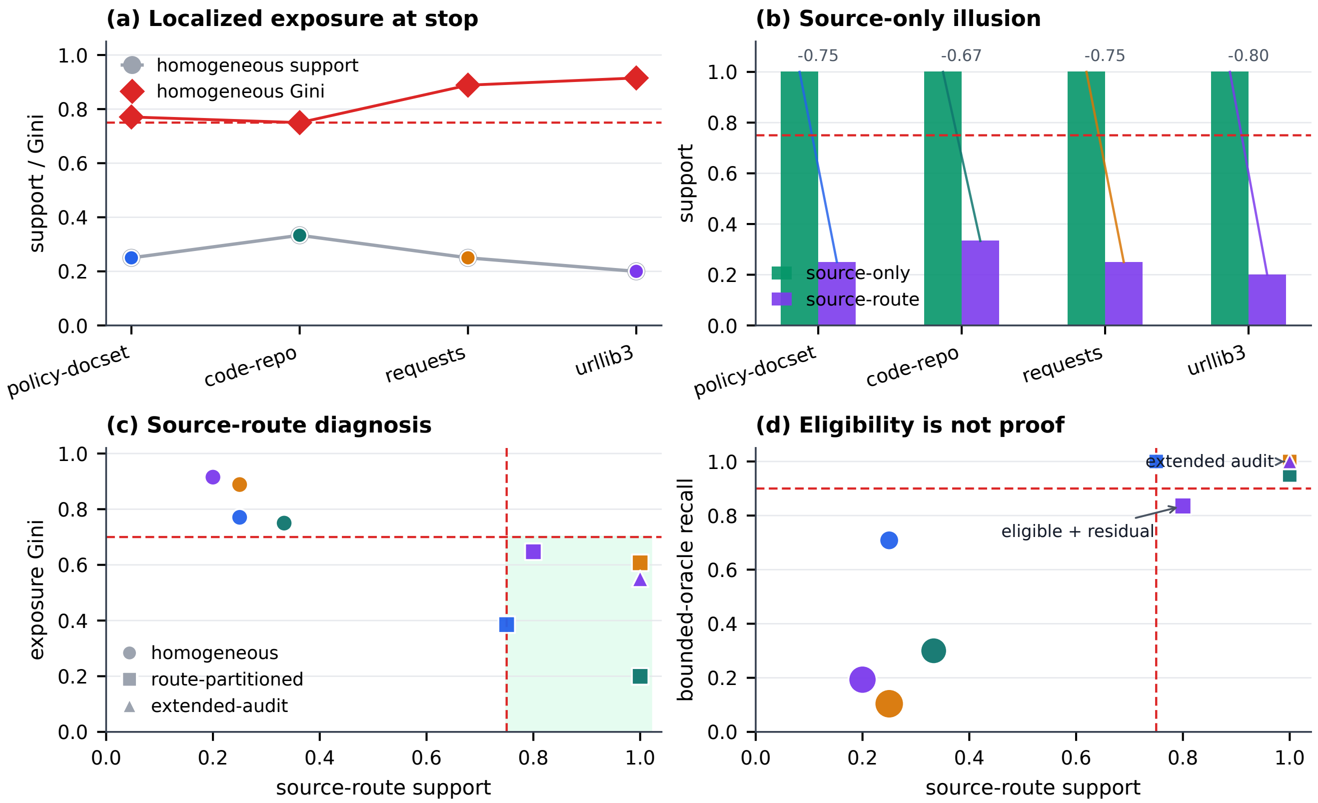

Figure 3: Diagnostic chain and granularity ablation. Local homogeneous stops have low source-route support and high concentration; source-only support is misleadingly complete; source-route support/Gini reveal the mismatch; and the `urllib3` route-partitioned state is eligible but residual-positive, with recall 0.835 below the post-hoc 0.90 threshold until extended audit reaches 1.000.

Table 4: Small claim-verification pilot with human-style audit routes. Routes are support, contradiction, exception-path, config-default, and scope-boundary. This deterministic pilot is a scoped sanity check for completion control, not a human-annotated benchmark. S/C/A reports SAFE/CONTINUE/ABSTAIN counts.

Policy	Unsafe	Safe	Unsafe S/C/A	Safe S/C/A	FCR	Safe cov.	Cost
Naive stop	4	2	4/0/0	2/0/0	1.000	1.000	0.0
Source-only	4	2	4/0/0	2/0/0	1.000	1.000	0.0
Verifier-gate	4	2	0/0/4	1/0/1	0.000	0.500	0.0
Eligibility-only	4	2	1/0/3	2/0/0	0.250	1.000	0.0
Full controller	4	2	0/4/0	2/0/0	0.000	1.000	3.5

only a precondition, and repair distinguishes unsafe residual states from broad complete states.

Table 5: Sensitivity sweep numeric summary. Ranges aggregate threshold settings or repair-budget settings within each external task.

Task	Sweep	SAFE rate	CONTINUE rate	ABSTAIN rate	Max FCR	Mean repair cost
`requests`	threshold	0.000-0.000	0.995-1.000	0.000-0.005	0.000	2854.6
`urllib3`	threshold	0.000-0.000	0.990-1.000	0.000-0.010	0.000	4831.8
`requests`	budget	0.000-0.000	0.635-1.000	0.000-0.365	0.000	3179.0
`urllib3`	budget	0.000-0.000	0.640-1.000	0.000-0.360	0.000	4323.5

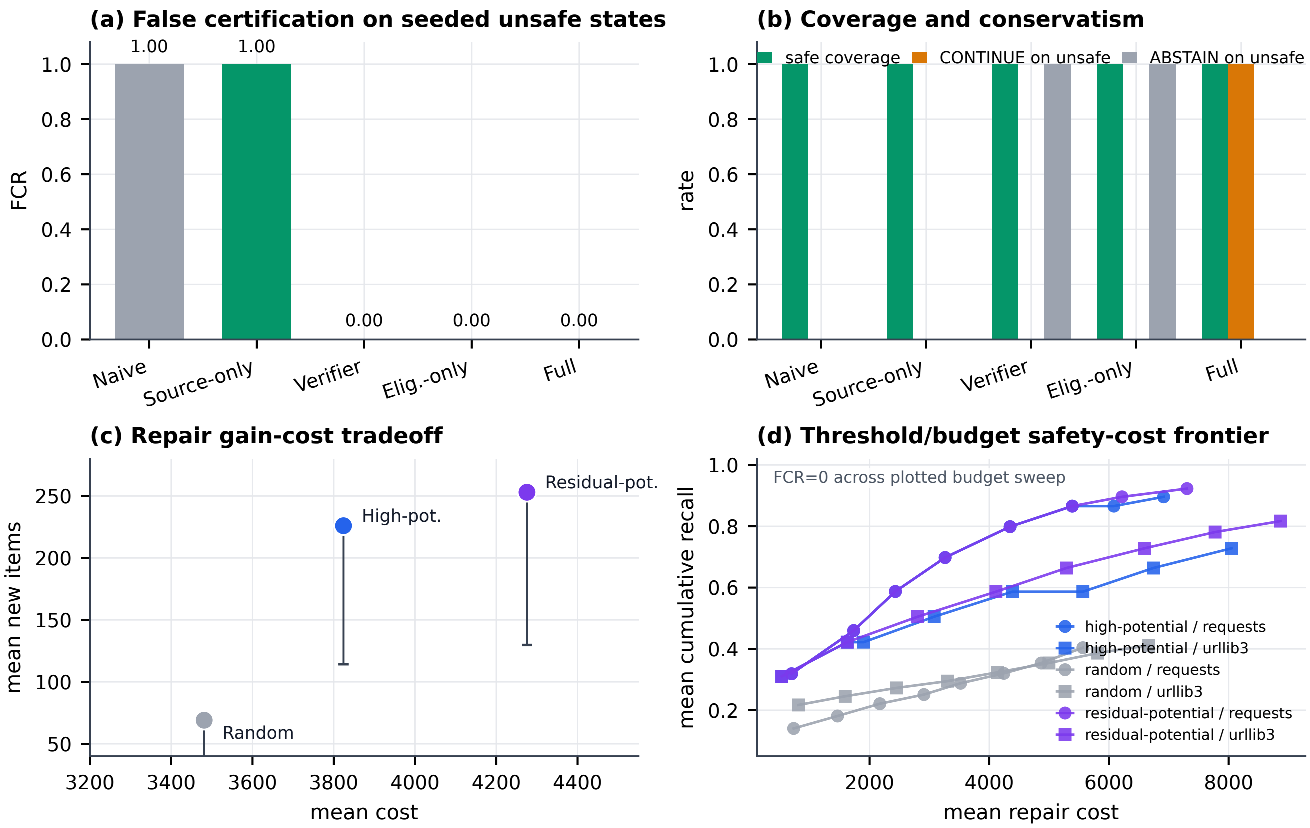

Figure 4: Controller evaluation dashboard. Decision-rule variants are scored on the same seeded unsafe repair states and seeded complete-state perturbations; the Verifier-gate baseline uses only unresolved/residual warnings and no source-route geometry. Repair-target variants keep the full decision rule fixed. The frontier panel shows how budget choices trade post-stop cost for cumulative recall while keeping FCR at zero in this sweep. Table 2 isolates the eligible-but-residual-positive boundary case.

8 Discussion

The experiments support a bounded certificate claim rather than an open-world guarantee. A SAFE decision means that the declared source-route condition is eligible and that budgeted repair did not reveal residual evidence; it does not mean that the world contains no further facts or that future searches would fail.

What the validation establishes. The ablations close the intended loop: homogeneous route reuse induces unsafe local stop evidence; source-route exposure detects the mismatch that source-only coverage misses; the `urllib3` boundary shows that eligibility is only a precondition; and repair separates broad-but-productive states from clean complete states.

Practical use of the controller. In deployment, support and Gini are stop-time warning signals: they indicate whether the evidence condition is broad enough to consider a certificate. Threshold and budget sweeps let an operator choose a more conservative or more permissive stopping posture before oracle scoring. Residual-potential repair then suggests where the next audit should look when the current certificate is weak. The output is an operational loop–SAFE when the declared condition is eligible and repair is nonproductive, CONTINUE when residual evidence is found, and ABSTAIN when the budget is exhausted without a defensible certificate.

Costs and boundaries. The controller trades convenience for conservative control. It can increase post-stop repair cost, and stricter thresholds or limited budgets can increase ABSTAIN decisions. Residual-potential is a mechanism-aligned heuristic, not an optimal active-search policy. The external repository oracles are pattern-defined over frozen snapshots, so they support mechanism validation but do not replace human semantic annotation. The claim-verification audit is a deterministic pilot rather than a human-annotated benchmark. These are the costs and boundaries of a bounded source-route completion-audit controller, not failures to deliver a universal completion guarantee.

Reproducibility and auditability. The implementation fixes snapshots, routes, seeds, thresholds, budgets, and oracle-scoring paths in config files. A three-seed config checks regeneration; the full seeded config produces the reported validation. Runtime-visible fields are separated from oracle-only fields, and stop states are fixed before scoring, so the controller cannot use the hidden workload it is supposed to audit.

9 Limitations

The external repository oracles are pattern-defined and therefore stronger than generated toy labels but weaker than human-annotated completion-audit benchmarks. They are nevertheless appropriate for this mechanism validation because the evaluation asks whether the controller accepts unsafe completion certificates, not whether it performs semantic vulnerability discovery. Source-route strata are task-designed, and other domains may need different route inventories or thresholds. Residual-potential repair is a mechanism-aligned probe, not an optimal active-search policy. The workflow pilot is an interface validation rather than a scored claim-verification benchmark; a human-annotated completion-claim audit remains the key external-validity test.

10 Conclusion

Local evidence does not automatically support a scope-wide completion claim. Evidence-condition geometry makes this conditioning explicit: the controller rejects unsafe local stops, repairs weak plausible strata, and abstains when no defensible certificate can be formed. Within bounded source-route completion-audit workflows, completion certificates should be judged by evidence condition, not stop signals alone.

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